CS50

Programming basiacs D2

C++

Outlines:

1- Loops.

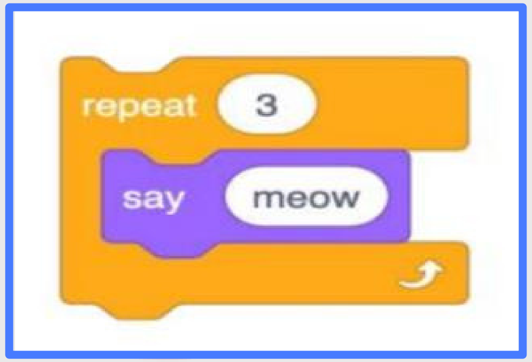
2- Functions.

3- Abstraction.

4- Operators and Types.

Loops :

Scratch



We can also utilize the loops building block from Scratch inour C++ programs. Try this code in your text editor:

```cpp
#include <iostream>
using namespace std;

int main(){
    cout << "Hello" << endl;
    cout << "Hello" << endl;
    cout << "Hello" << endl;
    return 0;
}
```

Notice this does as intended but has an opportunity for better design.

C++

```cpp
int counter = 3;
while(counter > 0){
    cout << "Hello" << endl;
    counter--;
}
```

Loops :

Count down

```cpp
#include <iostream>
using namespace std;

int main(){
    int i = 3;
    while(i > 0){
        cout << "Hello" << endl;
        i--;
    }
    return 0;
}
```

Count up

```cpp
#include <iostream>
using namespace std;

int main(){
    int i = 1;
    while(i <= 3){
        cout << "Hello" << endl;
        i++;
    }
    return 0;
}
```

Loops :

While loop

```
#include <iostream>
using namespace std;

int main(){
    int i = 1;
    while(i <= 3){
        cout << "Hello" << endl;
        i++;
    }
    return 0;
}
```

For loop

```
#include <iostream>
using namespace std;

int main(){
    for(int i = 1; i <= 3; i++){
        cout << "Hello" << endl;
    }
    return 0;
}
```

*Notice that the for loop includes three arguments. The first argument **int i = 0** starts our counter at zero. The second argument **i < 3** is the condition that is being checked. Finally, the argument **i++** tells the loop to increment by one each time the loop runs.*

Loops :

Do While loop

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    do
    {
        cin >> n;
    } while (n < 1);

    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            cout << "#";
        }
        cout << endl;
    }
}
```

Loops :

1. Given a number **N** print a line consisting of **N** hashes.

Examples:

Input:
5
Output:
#####

Loops :

1. Given a number **N** print a line consisting of **N** hashes.

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;
    for (int i = 0; i < n; i++)
    {
        cout << "#";
    }
    cout << endl;
}
```

Loops :

2. Given number **N** print numbers from 1 to N.

Examples:

Input:
5
Output:
1 2 3 4 5

Loops :

2. Given number **N** print numbers from 1 to N.

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;
    for (int i = 1; i <= n; i++)
    {
        cout << i << " ";
    }
    cout << endl;
}
```

Loops :

3. Given **N** numbers calculate it's summation.

Examples:

Input:
5 2 4 3 2 1
Output:
12

Loops :

3. Given **N** numbers calculate it's summation.

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;
    int sum = 0;
    for (int i = 1; i <= n; i++)
    {
        int x;
        cin >> x;
        sum += x;
    }
    cout << sum << endl;
}
```

Loops :

4. Given **N** print square of **#** of size N * N with N rows and N columns

Examples:

Input:
4
Output:
####

Loops :

4. Given **N** print square of **#** of size N * N with N rows and N columns

```cpp
#include <iostream>
using namespace std;

int main(){
    int n;
    cin >> n;

    for (int i = 0; i < n; i++){
        for (int j = 0; j < n; j++){
            cout << "#";
        }
        cout << endl;
    }
}
```

Loops:

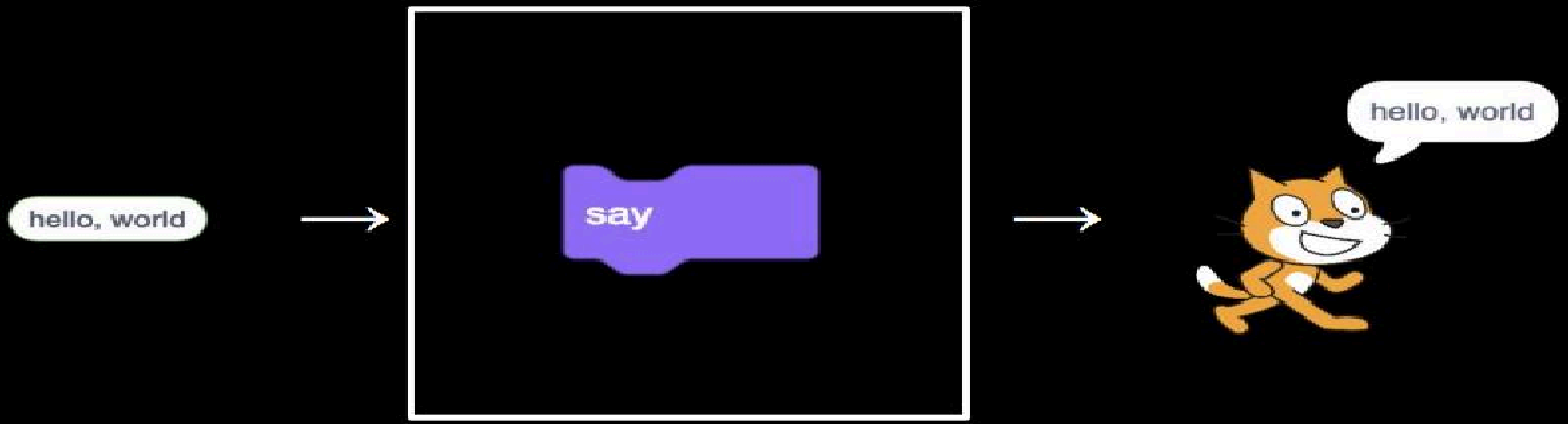
Given **N** print a pyramid of **#** on **N** lines

Input:
4
Output:
####

Input:
3
Output:
###

Functions:

arguments → function → side effects



Function :

```cpp
void print()
{
    cout << "Hello" << endl;
}
```

```cpp
int main(){
    for (int i = 0; i < 3; i++){
        print();
    }
}

```

```cpp
#include <iostream>
using namespace std;

void print();

int main(){
    for (int i = 0; i < 3; i++){

    }
}

void print(){
    cout << "Hello" << endl;
}
```

function prototype

function definition

Function :

function return type

function name

```
int add(int a, int b)
{
    int c = a + b;
    return c;
}
```

function first parameter

function second parameter

Functions :

Implement function that return the minimum number between two numbers.

Examples:

Input:
3 5
Output:
3

Functions :

Implement function that return the minimum number between two numbers.

```cpp
#include <iostream>
using namespace std;

int min(int a, int b){
    if(a <= b){
        return a;
    } else {
        return b;
    }
}

int main(){
    int x, y;
    cin >> x >> y;
    int ans = min(x, y);
    cout << ans << endl;
}
```

Functions :

1.Implement print function that print hello world

- call this function inside a loop
- move this loop to the function and pass to it an argument

Examples:

Input:
3
Output:
Hello, World **Hello, World** **Hello, World**

Functions :

1.Implement print function that print hello world

- call this function inside a loop
- move this loop to the function and pass to it an argument

```cpp
#include <iostream>
using namespace std;

void print(){
    cout << "Hello, World!" << endl;
}

int main(){
    int n;
    cin >> n;
    for (int i = 0; i < n; i++){
        print();
    }
}
```

Operators, overflow and type casting :

1.Given a Number ***N*** corresponding to a person's age (in days). Print his age in years, months and days, followed by its respective message "years", "months", "days".

Note:consider the whole year has **365** days and **30** days per month.

Examples:

Input:
400
Output:
1 years **1 months** **5 days**

Input:
800
Output:
2 years **2 months** **10 days**

Operators, overflow and type casting :

1.Given a Number ***N*** corresponding to a person's age (in days). Print his age in years, months and days, followed by its respective message "years", "months", "days".

Note: consider the whole year has **365** days and **30** days per month.

```
#include <iostream>
using namespace std;


int main(){
    int n;
    cin >> n;
    int years = n / 365;
    n %= 365;
    int months = n / 30;
    n %= 30;
    int days = n;
    cout << years << " years" << endl;
    cout << months << " months" << endl;
    cout << days << " days" << endl;
    return 0;
}
```

Operators, overflow and type casting :

Write a function that takes only one integer number and return the number is “Even” or “Odd”

Examples:

Input:
2
Output:
Even

Input:
5
Output:
Odd

Support
Team